

Feedback Solutions in Rowing Training

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ABSTRACT

The growing accessibility of wearables for sports encourages people to track their training progress and health information while exercising. The recorded information provides insights into the body and forms the basis for feedback systems. Though requirements for the feedback design vary greatly depending on the type of sport. Especially for activities involving the whole body, such as rowing. It trains strength and endurance, does not place excessive stress on leg joints, yet involves coordinating multiple muscles simultaneously, making it a complex sport by combining it all at once. In this paper, we look into research on feedback modalities and review the advantages and disadvantages of feedback examples in Human-Computer Interaction focusing on the rowing context.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *Ubiquitous and mobile computing*.

KEYWORDS

Rowing; feedback; physical activity; HCI for sports; wearables

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1 INTRODUCTION

Physical exercise is a vital part of our everyday life that helps the body function properly. It is responsible for our physical and mental health [1]. In recent years, people have been doing less sports [3] and excuse themselves from exercising due to lack of time, inconveniences, and environmental constraints [13, 21, 23]. Devices providing small home workouts [15] and exercise tracking [5] combat this trend by providing an engaging and fun environment to encourage participation. Some activities focus on specific muscle groups, e.g., running or cycling, while others engage the whole body, such as tennis, swimming, and rowing. The latter, for example, trains strength and endurance while not putting high workloads on the joints. Like other sports, without proper rowing technique, movements become

inefficient or even harmful, leading to injuries [4, 17]. The most common problem among rowers is back pain [8, 10, 11, 26]. To prevent this, coaches teach athletes the correct movements early in their training and provide feedback so that mistakes do not creep in [30]. In larger groups, trainers have to distribute their attention between multiple students, limiting space to recognise error in individuals. The current state of the art of technology supervision for rowing mainly uses wearables, such as smartwatches, speed coaches, etc. Those solutions do not provide any feedback about rowing techniques, so this task remains with the coach. However, in related research, there are sensor-based solutions that record the rower's movement, muscle activity, or acceleration to determine performance and generate feedback. IMU sensors, for example, are commonly used to deviate the exact position from the acceleration and orientation [6, 7, 9, 14]. In combination with proximity sensors, they determine the position in relation to the boat [19]. Electromyography (EMG) can provide insights into which muscles are active during a stroke, detecting use of proper technique [9]. Usually, wearable sensor-based solutions can cover one aspect of an activity, and combining several devices is necessary to capture the whole picture, making it inconvenient for the user. One solution is video-based recordings; for example, to analyse similarity in rowers' postures for an optimised team performance [2]. However, devices for a detailed whole-body recording often require a complex or cost-intensive setup and appropriate algorithms, making it less accessible for amateur sports or training on the water. Moreover, other projects focus on different types of feedback during training sessions to accommodate the diverse personal needs of athletes and challenging environments, like tactile feedback [25, 27], body-worn sensors and tools [12, 16, 28], and fitness trackers [18, 29]. Some of the developed systems concentrate on visual information with different levels of complexity, from LEDs [19] to fully rendered images [22]. Others use auditory feedback [19, 22], which reduces clutter in the field of view and makes it easier to process the information. Another approach presents haptic feedback, which makes a significant difference in gym training compared to classical ergometers, as it allows the creation of proper movement habits from the start [20, 22].

In this work, we use related work and an interview with a rowing instructor to describe the special requirements for feedback in the rowing context. Further, we present four developed solutions and discuss their advantages and disadvantages in relation to proper rowing techniques.

2 BACKGROUND

Here, we discuss movement techniques specific to rowing, as well as related work on movement and posture recording sensors. We



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further conduct an interview with a rower coach to identify common technique problems and how they can be addressed by the coach. These insights inform our categorisations of good and bad feedback examples (Section 3).

2.1 Related Work: The Rowing Technique

Rowing is a versatile sport in which athletes balance strength and the correct technique. This makes it a suitable exercise for the body and mind, directly improving body strength, endurance, and mental resilience. Rowing consists of repetitive movements consisting of two phases: drive and recovery. The drive is the movement's first part and is often called 'catch'. During this phase, the rower pulls the oars to accelerate the boat. In the recovery phase, the rower returns to the starting position and recovers from the drive strain. In rowing, one distinguishes between two boat styles: sculling, where each athlete has two oars, and sweep rowing, with only one oar per person. Although the basic principle is the same for both, it differs in the finish of the catch, moving the boat forward through the strength of the entire body and the leverage effect. During the drive, the power originates from the legs pushing, and the torso and arms pulling the oar. This exerts great force on the spine, leading to health problems if the technique is not used correctly [24].

2.2 Interview of a Teacher's Perspective

We interviewed a rowing coach and discussed the most common problems and techniques to gather further insights into the training process. He has been rowing for 15 years, with six years of coaching experience. The first common mistake is not following the proper order of movements, which is critical for rowing efficiency, balance, and overall performance. New athletes are pulling either their arms or legs too early, which causes sub-optimal force distribution during catch and recovery. They also bend the knees too much or do not keep their arms straight in the expected moments. The second mistake relates to poor posture, adding additional strain on the spine. Those problems are usually caused by unnecessary looking around or down, which, due to the bio-mechanical properties of muscles and bones, automatically changes forces inside the body. One way to correct those problems is training isolated movements and then connecting them gradually until they have internalised the full catch technique. As for all motor skills, the movement needs to be performed in the correct order but also has to become intuitive as during competitions there is no time to consciously check your technique.

3 FEEDBACK EXAMPLES

3.1 Good Examples

Rahman and Bakar [19] developed a system for controlling rowing techniques on an ergometer using ultrasonic and infrared sensors. During the drive and recovery phases, they monitored rowers' movements at the start, middle, and end positions. Then, they provided feedback via visual cues with LEDs near the ergometer screen and auditory buzzers along the slider. This solution monitors the entire rowing cycle and provides real-time feedback allowing athletes to correct their mistakes and concentrate on the amended technique. Moreover, placing LEDs next to the machine screen is

crucial to avoid the common mistake of looking down. It makes rowers maintain proper head position, which ultimately helps create good habits and improves balance in the boat; using buzzers allows rowers to process more information subconsciously. By providing feedback of solely the necessary information, these solutions teach rowers to learn the feeling of the correct movement and perform them intuitively.

Rauter et al. [20] created a space that simulates an outdoor environment, i.e., haptic (water interaction with oar), visual, and auditory information that are present while training on the water. To imitate oar interaction with water, they used a tendon-based parallel robot (TBPR), which can provide assistance or present a challenge depending on the athlete's ability level. The use of an actual boat setup allows for training specific movements that are required during sweep rowing (e.g., the need to rotate the body in one direction while maintaining balance). The ability to adjust the function of the TBPR allows the same machine to be used for both beginners and experts training. This adaptability addresses the different needs of these two groups: new athletes, who might lack proper technique and for whom any disturbance during exercise increases the likelihood of learning incorrect methods, and advanced rowers, who already know the correct stroke techniques and need challenges to train their muscle memory in difficult environments.

3.2 Bad Examples

Hohmuth et al. [9] describe a measurement system that approaches technique monitoring through the observation of body position during training sessions. With the usage of pressure, IMU, GPS and EMG sensors, this setup is able to record data that reveals boat movement and athlete progress. However, all of this information is displayed on the screen that is positioned just above the shoes holder in the boat. This, firstly, causes rowers to look down onto the screen, and secondly, changes the balance of the boat. While the latter can be overcome with additional weights on the other side of the boat (though it adds even more not-needed ballast), the first thing is a major problem in a sport where athlete posture plays a crucial role in the final result. While looking down onto the shoes level, the whole spine becomes more curved and shoulders are rolled to the front, limiting the range of motion during strokes. Consequently, this increases the risk of back problems and lower efficiency in rowing.

Sigrist et al. [22] recreated the TBPR system as described in [20] with additional improvements to the environmental setup. They installed three screens around the machine, and rendered videos of realistic water surroundings. A special algorithm followed the theoretical blade of the oar, its position, and interaction with water. They visualised the target and actual paths of the blade on the side screen in the place where they would be in the real world. This method provides immediate information about the differences between the desired and existing techniques, allowing for necessary corrections. However, the placement of this feedback on the side screen makes the athletes change their head position, which causes a number of changes in the bio-mechanical paths of muscles, resulting in an increased probability of pain. Moreover, this type of movement interrupts the balance of the whole body, resulting in

lower stroke efficiency, as well as an increased possibility of falling into the water.

4 DISCUSSION & CONCLUSION

The presented solutions introduce various approaches providing feedback in rowing, addressing different aspects of the sport. Designing feedback for this sport is challenging due to the environmental requirements. Additionally, a rigorous technique to optimise each stroke and preserve athletes' health, makes presenting real-time information challenging; and arguably, it is impossible to choose only one modality. First, the feedback's effectiveness depends on individual preferences, as well as proficiency level. Second, during different training phases, different senses are either overloaded or favoured. Moreover, it is important to remember that the system setup must be easy to use; otherwise, rowers will stop using it. For example, the idea presented by Hohmuth et al. [9] allows accurate movement detection. However, the display placement prevents athletes from keeping correct posture in favour of looking down at the screen. In order to correct this design, the feedback should be presented on the eye level. However, as the boat environment is not allowing for this placement (disturbed balance), an other solution might be augmented reality. Not only would it help with the proper technique, but it also removes additional weight. Another study uses visual feedback on the side screens and compares the desired to the actual blade path [22]. Although the feedback led to significant reduction of errors, the cue placement is sub-optimal as it provokes posture change and looking sideways. This method can induce mistakes in the rowing technique, affect the boat balance, and lead to pain in the long term. Our proposed solution is to move the visualisation to the front screen, allowing athletes to maintain proper posture and viewing direction.

The 'one size fits all' notion does not apply when designing feedback for a particular sport. It would be most beneficial to provide athletes with systems that use multimodal arrangements, allowing them to adjust to their individual needs. As many designs rely on visual information, augmented reality glasses may be one of the best methods to provide feedback to rowers, allowing them to maintain correct technique while observing related live information.

5 RESEARCHER BACKGROUND

Joanna Sorysz is a PhD student at RPTU Kaiserslautern-Landau and DFKI Germany. She is working on embedded intelligent technology in sports and physical therapy scenarios, focusing on affordable solutions for public use. Moreover, she is also interested in new devices for firefighting or EMT work. She draws knowledge about needed devices from her own experiences as an athlete in sports such as rowing, horse riding, and martial arts.

Meagan B. Loerakker is a PhD student in the Research Unit of Human Computer Interaction at TU Wien's Institute of Visual Computing and Human-Centered Technology. Her research focuses on how to design feedback for sports technologies based on proficiency level, but she also conducts research on human-centered artificial intelligence and technologies for well-being. She has been playing tennis for nearly 20 years, and occasionally plays volleyball and padel.

Bettina Eska is a PhD student at LMU Munich, where she researches sensor-based feedback systems for HCI in sports. She is also interested in designing and generating appropriate feedback for scenarios that require fine motor skills, not limited to sports. In feedback design, she focuses on feedback that encourages the user to reflect on the effects of the movement in order to foster body awareness.

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